Fall 2025

Problem 1 (Fa25). Suppose that a and b are real constants. Find all smooth bounded functions u from $\mathbb{R}^2$ to $\mathbb{R}$ satisfying $au_x + bu_y = u$.

Problem 2 (Fa25). Suppose that $f(x,t)$ with $x \in \mathbb{R}$ and $t \geqslant 0$ is a smooth solution to the heat equation on a circle, so that

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2} \qquad \text{and} \qquad f(x+1,t) = f(x,t)$$

Prove that the total heat $\displaystyle\int_0^1 f(x,t)\,\mathrm{d}x$ is conserved, in other words it does not depend on t.

Problem 3 (Fa25). The Bernoulli polynomials $B_n(x)$ can be defined by the properties

$$\frac{\mathrm{d}}{\mathrm{d}x} B_n(x) = nB_{n-1}(x) \qquad B_0(x) = 1 \qquad \int_0^1 B_n(x)\,\mathrm{d}x = 0 \text{ if } n > 0$$

- Use these properties to calculate $f(x,t) = \displaystyle\sum_{n \geqslant 0} B_n(x) \frac{t^n}{n!}$.
- Find the radius of convergence of the power series $\displaystyle\sum_{n \geqslant 0} B_n(0) \frac{t^n}{n!}$.

Problem 4 (Fa25). Prove that the function $\sum_{n>0} z^n/n^2$ for $|z| < 1$ can be analytically continued along any path starting from $1/2$ that does not meet 0 or 1. Explain why the point 0 has to be excluded.

Problem 5 (Fa25). The complex 2 by 2 matrices A, B, C satisfy $AB = BA$, $AC = CA$, $BC \neq CB$. Prove that A is a multiple of the identity matrix.

Problem 6 (Sp20, Fa20, Fa25). Let V be a vector space of dimension n over a finite field with q elements. Prove that the number of subspaces $W \subseteq V$ of dimension k is equal to

$$\frac{\prod_{j=1}^{n} (q^j - 1)}{\left(\prod_{j=1}^{k} (q^j - 1)\right)\left(\prod_{j=1}^{n-k} (q^j - 1)\right)}$$

Problem 7 (Fa25). Give examples of two nonisomorphic non-abelian groups of order 8, and explain why they are not isomorphic.

Problem 8 (Fa25). How many abelian groups are there with 1000000 elements?

Problem 9 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25). For each $(a, b, c) \in \mathbb{R}^3$, consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of (a, b, c) for which the series converges absolutely; converges but not absolutely; diverges.

Problem 10 (Fa25). If $C \subseteq [0, 1]$ is the Cantor set, determine $C + C$, the set of all numbers that are the sum of 2 elements of C.

Problem 11 (Fa25). Show how to calculate the integral $\displaystyle\int_{-\infty}^{\infty} e^{2\pi ixy-\pi x^2}\,\mathrm{d}x$.

Problem 12 (Fa25). Prove that any holomorphic automorphism of the Riemann sphere is a fractional linear transformation $z \mapsto az + b/cz + d$.

Problem 13 (Fa25). Legendre polynomials $P_n(x)$ are defined inductively as real polynomials of degree exactly n such that

$$\int_{-1}^{1} P_m(x)P_n(x)\,\mathrm{d}x = 0 \quad \text{for all} \quad m < n$$

Prove that P_n has n simple zeros in the interval $[-1, 1]$.

Problem 14 (Fa25). What is the maximal dimension of a linear subspace in $\mathbb{R}^{2n+1}$ on which the quadratic form $x_0^2 + x_1x_2 + x_3x_4 + \cdots + x_{2n-1}x_{2n}$ takes only nonpositive values?

Problem 15 (Fa25). Ramanujan's taxicab number $1729 = 10^3+9^3 = 12^3+1^3$ is the smallest integer expressible as the sum of two positive cubes in two different ways. Prove that

$$x^{1729} \equiv x \pmod{1729}$$

for all integers x.

Problem 16 (Fa25). How many rectangles in the plane are there with sides parallel to the coordinate axes such that all coordinates of all corners are single digit integers 0, 1, 2,…, 9?